CS470: Introduction to Artificial Intelligence

인공지능개론

(Syllabus version 0.1 – 2025/12/31)

Instructor

Dr. Hyunwoo J. Kim (hyunwoojkim@kaist.ac.kr)
Office hour: Anytime by appointment

Class Hour & Location

Class hour: TTh 10:30AM - 12:00PM
Classroom (E3-1) Information & Electronics B/D (정보전자공학동 (1501호) 강의실-제1공동강의실)

Teaching Assistant

TBD
Office hour: Anytime by appointment

Description

This is an introductory course on Artificial Intelligence (AI) for undergraduate students. The goal is to acquire fundamental knowledge of intelligence, agents, reasoning, and inference. To do that, this course overviews the topics from classic AI methodologies to modern machine-learning techniques, covering techniques of searching, constraint satisfaction, logical reasoning, planning, and intelligent robot to provide a comprehensive spectrum of AI. Please, note that we will NOT cover the entire range of AI areas. The course aims to meet different levels of demands from diverse groups of audiences by providing (1) recent trends of AI for beginners, (2) fundamental theories of AI for those who wish to use techniques, and (3) substantial practice for problem-solving and applications.

Reference Materials:

AIMA book: Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach (4th ed.)*, Pearson Education, 2021.

Prerequisites:

Linear Algebra, Statistics, and data structures. You must be comfortable with python programming, Linux, and mathematical notation.
It is recommended to take CS376 <machine learning> before or with this class.

Schedule (Tentative)

# Week	Topics and slides	Assignment /Project / ETC (TBD)
1	Introduction	
2	Uninformed Search	
3	Informed Search	
4	Game trees: Adversarial Search	
5	Uncertainty and Utilities	
6	Markov Decision Process I	
7	Markov Decision Process II	
8	Midterm	
9	Reinforcement Learning I	
10	Reinforcement Learning II	
11	LLM-based Agents	
12	Neural Networks I	
13	Neural Networks II	
14	Large-scale AI systems (scaling laws)	
15	Advanced Topics in AI	
16	Final Exam	

Grading Criteria

- Assignments & projects (25%)
- Attendance & class participation (5%)
- Midterm exam (30%)
- Final exam (40%)

Assignment Policy

The assignments include written work, essays, implementation, and analysis.

Each assignment is to be completed by you alone. You should not share the code with other students or other people. Note that this class does **not allow publishing any provided class content** in public (e.g., GitHub or blog).

Late Submission Policy:
Assignments submitted after the due time will incur a 20% penalty per day as follows.

Submission Time After Due	Penalty Applied	Maximum Score Possible
On time	0%	100%
Up to 24 hours late	20%	80%
1–2 days late	40%	60%
2–3 days late	60%	40%
3–4 days late	80%	20%
5 days or more late	100%	0%